JNTUA R23 Regulations

CHEMISTRY LAB

(23A51202P)

I Year B.Tech. – Artificial Intelligence & Machine Learning

II Semester

Common to EEE, ECE, CSE, IT & allied branches

L T P C = 0 0 2 1

LAB RECORD

Name : _____

Roll No. : _____

Branch / Section : _____

Academic Year : _____

Course Objectives & Outcomes

Course Objective:

- To verify the fundamental concepts with experiments.

Course At the end of the course, the students will be able to

1. Determine the cell constant and conductance of solutions.
2. Prepare advanced polymer Bakelite materials.
3. Measure the strength of an acid present in secondary batteries.
4. Analyse the IR spectra of some organic compounds.
5. Calculate strength of acid in Pb-Acid battery.

Reference:

"Vogel's Quantitative Chemical Analysis", 6th Edition, Pearson Publications, by J. Mendham, R.C. Denney, J.D. Barnes and B. Sivasankar.

INDEX

S.No	Name of the Experiment	Date	Page	Sign
1	Measurement of 10Dq by spectrophotometric method		4	
2	Conductometric titration of strong acid vs. strong base		7	
3	Conductometric titration of weak acid vs. strong base		9	
4	Determination of cell constant and conductance of solutions		11	
5	Potentiometry - determination of redox potentials and emfs		13	
6	Determination of strength of an acid in Pb-Acid battery		15	
7	Preparation of a Bakelite		17	
8	Verification of Lambert-Beer's law		19	
9	Wavelength measurement of a sample through UV-Visible spectroscopy		21	
10	Identification of simple organic compounds by IR		23	
11	Preparation of nanomaterials by precipitation method		25	
12	Estimation of ferrous iron by dichrometry		27	

Experiment No. 1

MEASUREMENT OF 10Dq BY SPECTROPHOTOMETRIC METHOD

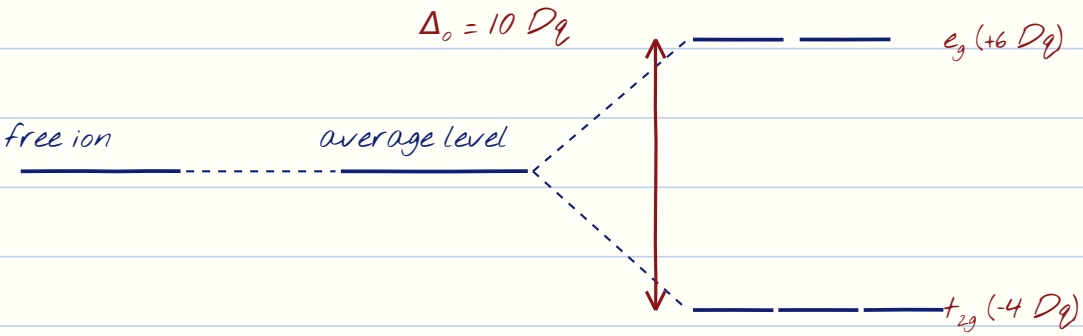
Aim: To determine the crystal field splitting energy (10Dq) of a transition metal complex, $[Cu(H_2O)_6]^{2+}$, using a UV-Visible spectrophotometer.

Apparatus: UV-Visible spectrophotometer, cuvettes, volumetric flasks (50 mL), pipettes, beaker, wash bottle.

Chemicals: $CuSO_4.5H_2O$ solution (0.1 M), distilled water.

Principle:

In an octahedral complex the five degenerate d-orbitals split into two sets: lower t_{2g} (d_{xy}, d_{yz}, d_{zx}) and higher e_g ($d_{x^2-y^2}$, d_{z^2}). The energy gap between them is the crystal field splitting energy, 10Dq (Δ_o). When the complex absorbs visible light, an electron is promoted from t_{2g} to e_g; the wavelength of maximum absorption (λ_{max}) gives 10Dq.



Octahedral field (six H_2O ligands)

$$10\,Dq\,(cm^{-1}) = 1 / \lambda_{max}\,(cm) = 10^7 / \lambda_{max}\,(nm)$$

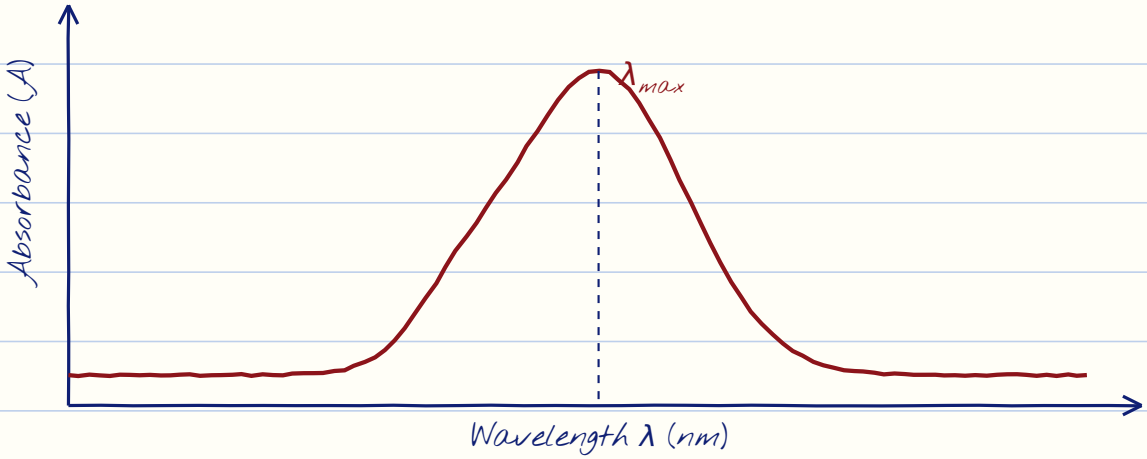
$$10\,Dq\,(KJ\,mol^{-1}) = 10\,Dq\,(cm^{-1}) \times 0.01196$$

Procedure:

1. Prepare the given complex solution of suitable concentration in distilled water.
2. Switch on the spectrophotometer and allow it to warm up. Fill one cuvette with distilled water (blank) and set $A = 0$ at each wavelength.
3. Rinse the other cuvette with the sample solution, fill it and place it in the sample holder.
4. Record the absorbance of the sample from 400 nm to 900 nm at intervals of 10–20 nm.
5. Plot absorbance (A) vs wavelength (λ) and find λ_{max} from the peak.
6. Calculate 10Dq from λ_{max}.

Observation table:

λ (nm)	Absorbance (A)	λ (nm)	Absorbance (A)



Calculation: λ_{max} = _____ nm → $10Dq = 10^7 / \lambda_{max}$ = _____ cm^{-1} = _____ $KJ\ mol^{-1}$

Result: The crystal field splitting energy (10Dq) of the complex = _____ cm^{-1}.

Experiment No. 2

CONDUCTOMETRIC TITRATION OF STRONG ACID vs STRONG BASE

Aim: To determine the strength of a strong acid (HCl) by titrating it against a strong base (NaOH) conductometrically.

Apparatus: Conductivity meter with dipping cell, burette, pipette, 250 mL beaker, magnetic stirrer / glass rod, burette stand.

Chemicals: HCl solution (unknown), 0.1 N NaOH solution, distilled water.

Principle:

Conductance depends on the number and mobility of ions. H^+ ions have very high mobility. When NaOH is added to HCl, fast-moving H^+ ions are replaced by slow-moving Na^+ ions, so conductance falls steadily until the equivalence point. After the end point, extra OH^- ions (also highly mobile) cause conductance to rise sharply. The intersection of the two lines gives the end point.

$$H^+ + Cl^- + Na^+ + OH^- \rightarrow Na^+ + Cl^- + H_2O$$

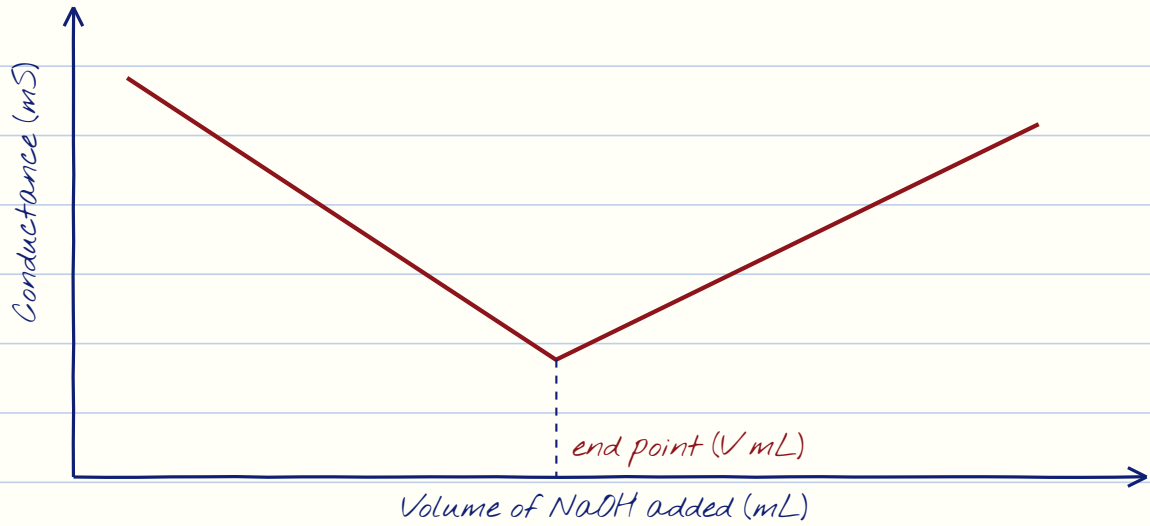
Procedure:

1. Rinse and fill the burette with 0.1 N NaOH.
2. Pipette out 20 mL of the given HCl into a clean beaker and add about 30 mL distilled water so that the electrode is well immersed.
3. Dip the conductivity cell in the solution and note the initial conductance.
4. Add NaOH in 1 mL increments, stir well and note the conductance after every addition.
5. Continue until conductance has increased well beyond the minimum (about 8–10 readings after the end point).

6. Plot conductance (y-axis) vs volume of NaOH (x-axis). Extrapolate the two lines; their intersection gives the end-point volume.

Observation table:

Vol. of NaOH (mL)	Conductance (mS)	Vol. of NaOH (mL)	Conductance (mS)



Calculation:

$$N_1V_1 = N_2V_2 \; (N_1 = N_2V_2 \,/\, V_1)$$

V_1 = volume of HCl = 20 mL, N_2 = 0.1 N, V_2 = end-point volume of NaOH = ____ mL

Normality of HCl, N_1 = _____ N; Strength = N_1 x 36.5 = _____ g/L

Result: Strength of the given HCl = _____ N (_____ g/L).

Experiment No. 3

CONDUCTOMETRIC TITRATION OF WEAK ACID vs STRONG BASE

Aim: To determine the strength of a weak acid (CH_3COOH) by titrating it against a strong base ($NaOH$) conductometrically.

Apparatus: Conductivity meter and cell, burette, pipette, beaker, stirrer, burette stand.

Chemicals: Acetic acid solution (unknown), 0.1 N $NaOH$, distilled water.

Principle:

Acetic acid is weakly ionised, so its initial conductance is low. On adding $NaOH$, sodium acetate (a strong electrolyte) is formed and the conductance rises slowly (a slight initial dip may occur because the common ion CH_3COO^- suppresses ionisation of the acid). After the equivalence point, free OH^- ions from excess $NaOH$ make the conductance rise steeply. The break in the graph is the end point.

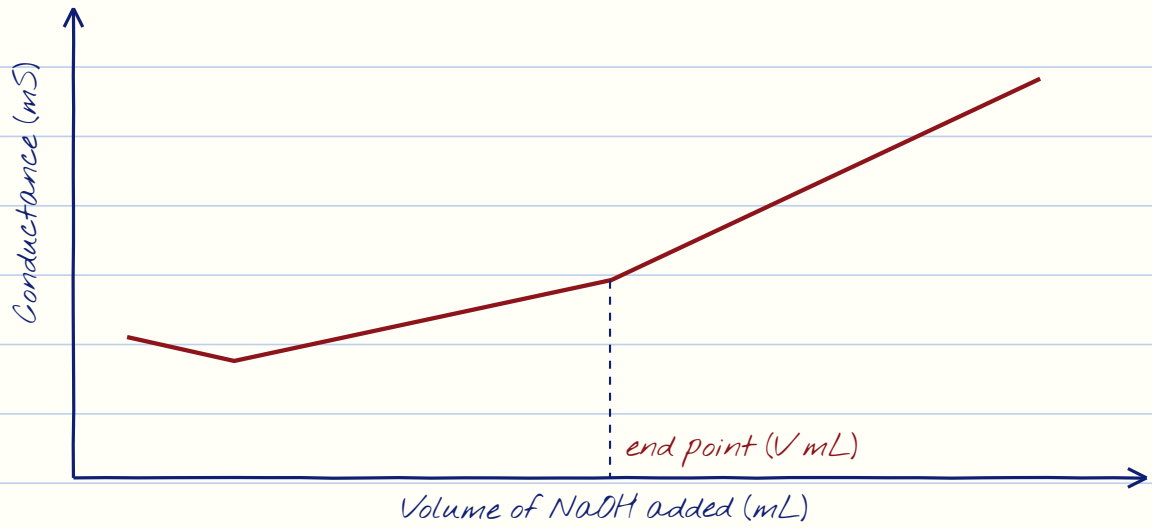
$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$

Procedure:

1. Fill the burette with 0.1 N $NaOH$.
2. Pipette 20 mL of acetic acid into a beaker, add about 30 mL distilled water and immerse the conductivity cell.
3. Note the initial conductance, then add $NaOH$ 1 mL at a time, stirring, and note the conductance each time.
4. Continue for 8–10 readings beyond the end point.
5. Plot conductance vs volume of $NaOH$; the intersection of the two straight portions gives the end-point volume.

Observation table:

Vol. of NaOH (mL)	Conductance (mS)	Vol. of NaOH (mL)	Conductance (mS)



Calculation:

$$N_1V_1 = N_2V_2$$

N_1 (CH_3COOH) = N_2V_2 / V_1 = _____ N; Strength = $N_1 \times 60$ = _____ g/L

Result: Strength of the given acetic acid = _____ N (_____ g/L).

Experiment No. 4

DETERMINATION OF CELL CONSTANT AND CONDUCTANCE OF SOLUTIONS

Aim: To determine the cell constant of a conductivity cell using standard KCl solution and to find the conductance of given solutions.

Apparatus: Conductivity meter, conductivity cell, volumetric flasks, pipettes, beakers, thermometer.

Chemicals: 0.01 M KCl (standard), given solutions (e.g. NaCl, HCl), conductivity water.

Principle:

Conductance (G) of a solution between two electrodes of area a separated by a distance l is $G = \kappa (a / l)$, where κ is the specific conductance. The ratio l/a is the cell constant (K_{cell}) and depends only on the cell geometry. It is found by measuring G of a standard KCl solution whose κ is accurately known (0.01 M KCl: κ = 1.413 mS cm^{-1} at 25 °C).

$$K_{cell} = l / a = \kappa(KCl) / G(KCl) \ [cm^{-1}]$$

$$\kappa(\text{solution}) = K_{cell} \times G \qquad \Lambda_m = 1000\,\kappa / C \ [S\ cm^2\ mol^{-1}]$$

Procedure:

1. Rinse the cell with conductivity water, then with a little 0.01 M KCl.
2. Fill the cell with 0.01 M KCl so the electrodes are fully immersed, note temperature and the conductance G.
3. Calculate K_{cell} from the known κ of KCl.
4. Wash the cell and measure G for each of the given solutions of different concentrations.

5. Calculate specific conductance κ and molar conductivity Λ_m of each solution.

Observation table:

Solution	Conc. C (M)	G (mS)	$\kappa = K_{cell} \cdot G$	Λ_m
0.01 M KCl	0.01		1.413	

Calculation: $K_{cell} = 1.413 / G(KCl) =$ _____ cm^{-1}

Result: Cell constant of the conductivity cell = _____ cm^{-1}. Specific conductance and molar conductivity of the given solutions are tabulated above.

Experiment No. 5

POTENTIOMETRY – DETERMINATION OF REDOX POTENTIALS AND EMFs

Aim: To determine the electrode (redox) potentials of Zn and Cu half-cells and the emf of the Daniell cell potentiometrically.

Apparatus: Potentiometer / digital voltmeter, saturated calomel electrode (SCE), Zn and Cu electrodes, beakers, salt bridge ($KCl|KNO_3$-agar), connecting wires.

Chemicals: $ZnSO_4$ and $CuSO_4$ solutions (1 M, 0.1 M, 0.01 M).

Principle:

The potential of an electrode dipping in a solution of its own ions depends on ion concentration and is given by the Nernst equation. Electrode potential is measured against a reference electrode of known potential (SCE = 0.244 V). The emf of a cell is the difference of the two electrode potentials.

$$E = E^\circ + (0.0591 / n) \log [M^{n+}] \text{ (at 25 °C)}$$

$$E_{cell} = E_{cathode} - E_{anode}$$

$$Zn \mid Zn^{2+}(C_1) \parallel Cu^{2+}(C_2) \mid Cu \quad E^\circ_{cell} = 0.34 - (-0.76) = 1.10 \text{ V}$$

Procedure:

1. Connect the Zn electrode in $ZnSO_4$ solution to the SCE through a salt bridge, measure the emf (E_{cell}).
2. Calculate $E_{Zn} = E_{SCE} - E_{cell}$ (as per polarity). Repeat for different concentrations.
3. Similarly measure the emf of $Cu \mid CuSO_4$ against SCE and calculate $E_{Cu} = E_{cell} + E_{SCE}$.
4. Set up the Daniell cell $Zn \mid ZnSO_4 \parallel CuSO_4 \mid Cu$ using the salt bridge and measure its emf directly.
5. Compare the observed values with the Nernst-equation values.

Observation table:

Half-cell	Conc. (M)	emf vs SCE (V)	Electrode potential (V)

Calculation: E_{cell} (observed) = _____ V; E_{cell} (Nernst) = _____ V.

Result: E_{Zn} = _____ V, E_{Cu} = _____ V and the emf of the Daniell cell = _____ V.

Experiment No. 6

DETERMINATION OF STRENGTH OF AN ACID IN Pb-ACID BATTERY

Aim: To determine the strength of sulphuric acid present in a lead-acid (Pb-acid) battery by titration with standard NaOH.

Apparatus: Burette, pipette (10 mL), standard flask (100 mL), conical flask, burette stand, white tile, funnel.

Chemicals: Battery acid sample, 0.1 N NaOH (standard), phenolphthalein indicator, distilled water.

Principle:

Battery acid is dilute H_2SO_4. It is neutralised by standard NaOH (acid-base titration). Phenolphthalein is colourless in acid and turns pink in base, marking the end point.

$$H_2SO_4 + 2\ NaOH \rightarrow Na_2SO_4 + 2\ H_2O$$

Procedure:

1. Pipette 5 mL of battery acid into a 100 mL standard flask and make up to the mark with distilled water (dilution factor = 20).
2. Pipette 10 mL of the diluted acid into a conical flask and add 2 drops of phenolphthalein.
3. Titrate against 0.1 N NaOH from the burette until a permanent light-pink colour appears (end point).
4. Repeat until two concordant readings are obtained.

Observation table:

S.No	Vol. of acid (mL)	Burette reading (mL)	Vol. of NaOH (mL)
1	10		
2	10		
3	10		

Calculation:

$N_1V_1 = N_2V_2 \rightarrow N_1 = (0.1 \times V_2) / 10 =$ _____ N (diluted acid)

Normality of battery acid = $N_1 \times 20$ (dilution factor) = _____ N

Strength = Normality × Equivalent weight (49) = _____ g/L

Result: Strength of the acid in the given Pb-acid battery =

_____ N (_____ g/L of H_2SO_4).

Experiment No. 7

PREPARATION OF A BAKELITE

Aim: To prepare Bakelite (phenol-formaldehyde resin) in the laboratory.

Apparatus: Beaker (250 mL), glass rod, measuring cylinder, filter paper, funnel, hot-air oven / watch glass, weighing balance.

Chemicals: Phenol (5 g), formaldehyde (40 % formalin, 7–8 mL), conc. HCl (few drops, catalyst), distilled water.

Principle:

Bakelite is a thermosetting polymer formed by condensation polymerisation of phenol and formaldehyde. Phenol first reacts with formaldehyde at its ortho/para positions to give hydroxymethyl phenols, which condense (losing water) to give a linear polymer (novolac); further heating with formaldehyde cross-links it into a rigid 3-D network — Bakelite.

$$n\,C_6H_5OH + n\,HCHO \rightarrow [\,-C_6H_3(OH)-CH_2-\,]_n + n\,H_2O$$

Procedure:

1. Take 5 g of phenol in a clean beaker and add 7–8 mL of 40 % formaldehyde solution (carry out in a fume hood / well-ventilated area, wear gloves).
2. Add a few drops of conc. HCl carefully while stirring with a glass rod.
3. The reaction is exothermic; stir until the mixture turns viscous, pinkish and finally solidifies into a hard mass.
4. Allow it to cool, wash the solid with distilled water to remove unreacted acid and monomers, then dry it in the oven / air.
5. Weigh the dry Bakelite and calculate the yield.

Properties & uses:

- Hard, rigid, heat-resistant, good electrical insulator, and resistant to acids (not alkalis).
- Used for electrical switches, plugs, sockets, handles of utensils, telephone parts and laminates.

Result: Bakelite was prepared. Weight of Bakelite obtained = _____ g.

Experiment No. 8

VERIFICATION OF LAMBERT-BEER'S LAW

Aim: To verify Lambert-Beer's law using a colorimeter / spectrophotometer with $KMnO_4$ solutions of different concentrations.

Apparatus: Colorimeter or UV-Vis spectrophotometer, cuvettes, volumetric flasks (50 mL), burette / pipettes.

Chemicals: Standard $KMnO_4$ solution (stock, 0.001 M), distilled water.

Principle:

Beer's law states that absorbance of a solution is directly proportional to its concentration, and Lambert's law that it is proportional to the path length. Combined:

$$A = \log(I_0 / I_t) = \varepsilon c l$$

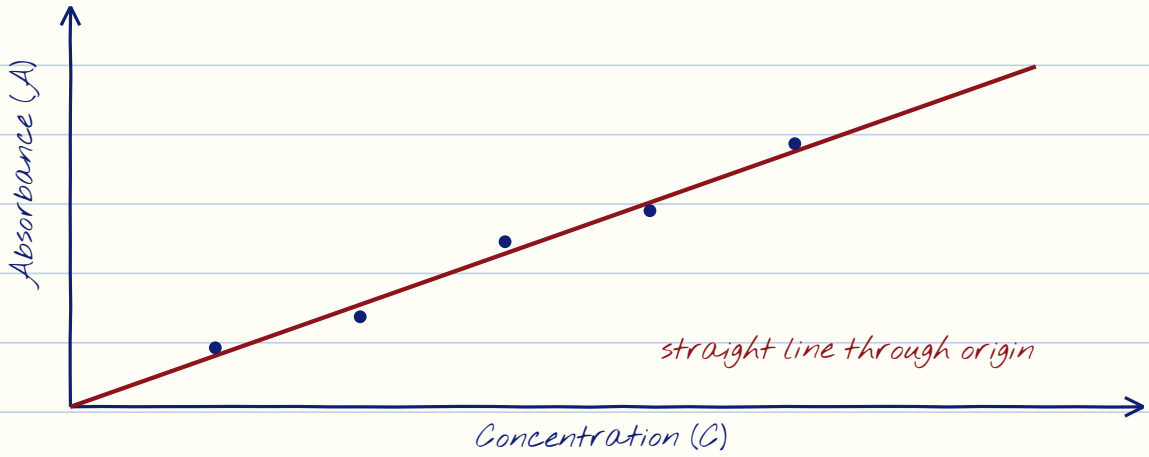
where A = absorbance, ε = molar absorptivity ($L\ mol^{-1}\ cm^{-1}$), c = concentration (mol/L), l = path length (cm). A plot of A vs c is a straight line through the origin if the law is obeyed.

Procedure:

1. Prepare 5–6 solutions of different concentrations by diluting the stock solution (e.g. 2, 4, 6, 8, 10 mL made up to 50 mL).
2. Determine λ_{max} ($\approx$ 525 nm for $KMnO_4$) and set the instrument at this wavelength.
3. Zero the instrument with distilled water (blank).
4. Measure the absorbance of each solution.
5. Plot absorbance (y) vs concentration (x). A straight line through the origin verifies the law.

Observation table:

S.No	Vol. of stock (mL)	Concentration (M)	Absorbance (A)
1			
2			
3			
4			
5			
6			



Calculation: Slope of the graph = εl = _____ ; ε = slope / l = _____ L mol^{-1} cm^{-1}

Result: A vs C is a straight line passing through the origin, hence Lambert-Beer's law is verified.

Experiment No. 9

WAVELENGTH MEASUREMENT OF SAMPLE THROUGH UV-VISIBLE SPECTROSCOPY

Aim: To determine the wavelength of maximum absorption (λ_{max}) of the given sample using a UV-Visible spectrophotometer.

Apparatus: UV-Visible spectrophotometer, quartz / glass cuvettes, volumetric flask, pipette.

Chemicals: Given sample solution (e.g. $KMnO_4$ / $CuSO_4$), solvent (distilled water).

Principle:

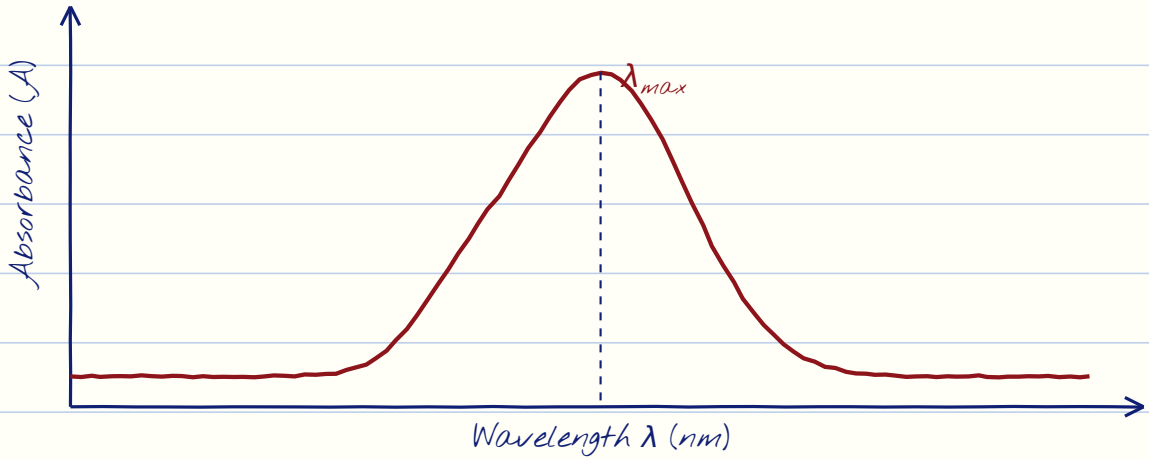
Molecules absorb UV (200–400 nm) or visible (400–800 nm) radiation and undergo electronic transitions (e.g. $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$, d–d). The absorbance is measured as a function of wavelength; the wavelength at which absorbance is maximum is λ_{max}, characteristic of the species. The energy of the transition is $E = hc / \lambda$.

Procedure:

1. Prepare a dilute solution of the sample in a suitable solvent.
2. Fill a cuvette with the pure solvent and use it as blank to set the baseline ($A = 0$).
3. Fill another cuvette with the sample and place it in the sample compartment.
4. Scan the spectrum over the required range (e.g. 300–800 nm) and record absorbance at regular wavelength intervals.
5. Plot A vs λ and note the wavelength of the peak (λ_{max}).

Observation table:

λ (nm)	Absorbance (A)	λ (nm)	Absorbance (A)



Result: λ_{max} of the given sample = _____ nm.

Experiment No. 10

IDENTIFICATION OF SIMPLE ORGANIC COMPOUNDS BY IR

Aim: To identify the functional groups present in simple organic compounds from their infrared (IR) spectra.

Apparatus: FT-IR spectrometer, NaCl / KBr plates, KBr pellet press, agate mortar, sample holder.

Chemicals: Given organic samples (liquid / solid), spectroscopic-grade KBr, acetone (for cleaning).

Principle:

A molecule absorbs IR radiation when the frequency matches the vibrational frequency of a bond and the vibration changes the dipole moment. Each functional group absorbs at a characteristic wavenumber (cm^{-1}), so the IR spectrum acts as a fingerprint for the group.

Characteristic IR absorptions:

Bond / group	Wavenumber (cm^{-1})	Nature
O–H (alcohol)	3200 – 3600	broad, strong
O–H (carboxylic acid)	2500 – 3300	very broad
N–H (amine / amide)	3300 – 3500	medium
C–H (alkane, sp^3)	2850 – 2960	strong
C–H (alkene / aromatic)	3000 – 3100	medium
C≡C / C≡N	2100 – 2260	weak / medium
C=O (aldehyde, ketone, acid)	1670 – 1740	strong, sharp
C=O (ester)	1735 – 1750	strong
C=C (alkene / aromatic)	1450 – 1680	variable
C–O	1000 – 1300	strong

Procedure:

1. Liquid sample: place a drop between two NaCl plates to form a thin film. Solid sample: grind ~1 mg with ~100 mg dry KBr and press into a transparent pellet.
2. Record the background spectrum and then the spectrum of the sample.
3. Note the major peaks (wavenumber, intensity) and assign them using the table.
4. Deduce the functional groups present and hence identify the compound.

Observation table:

Sample	Peak (cm^{-1})	Assignment / group

Result: The functional groups present in the given sample are _____ and the compound is identified as _____.

Experiment No. 11

PREPARATION OF NANOMATERIALS BY PRECIPITATION METHOD

Aim: To prepare zinc oxide (ZnO) nanoparticles by the chemical precipitation method.

Apparatus: Beakers, magnetic stirrer, burette / dropper, centrifuge or filtration set, hot-air oven, muffle furnace, crucible, weighing balance.

Chemicals: Zinc acetate dihydrate (0.1 M), NaOH (0.2 M), distilled water, ethanol.

Principle:

In the precipitation (bottom-up) method, a soluble metal salt is converted into an insoluble hydroxide by adding a base under controlled stirring, giving very fine particles. Thermal decomposition (calcination) of the hydroxide gives the metal oxide nanoparticles. Concentration, pH, temperature and stirring rate control the particle size.

$$Zn(CH_3COO)_2 + 2\ NaOH \rightarrow Zn(OH)_2 \downarrow + 2\ CH_3COONa$$

$$Zn(OH)_2 \xrightarrow{-(\Delta,\ \text{calcination})} ZnO + H_2O$$

Procedure:

1. Dissolve zinc acetate in distilled water (0.1 M, 50 mL) in a beaker and keep it on the magnetic stirrer.
2. Add 0.2 M NaOH solution dropwise with constant stirring until a white precipitate of $Zn(OH)_2$ forms (pH ≈ 10–12).
3. Continue stirring for about 2 hours to complete the reaction.
4. Separate the precipitate by centrifugation / filtration and wash it several times with distilled water and then ethanol.
5. Dry the precipitate in an oven at about 80–100 °C.

6. Calcine the dried powder in a muffle furnace (about 400–500 °C, 2 h) to get ZnO nanopowder; cool and weigh.

Characterisation particle size and structure are studied by XRD, SEMITEM, and optical band gap by UV-Vis.

Result: ZnO nanoparticles were prepared by precipitation method.

Weight of product obtained = _____ g.

Experiment No. 12

ESTIMATION OF FERROUS IRON BY DICHROMETRY

Aim: To estimate the amount of ferrous iron (Fe^{2+}) present in the given solution by titration against standard potassium dichromate ($K_2Cr_2O_7$).

Apparatus: Burette, pipette (20 mL), conical flask, standard flask (250 mL), burette stand, funnel, white tile.

Chemicals: Standard $K_2Cr_2O_7$ (0.1 N), ferrous solution (Mohr's salt / FAS), dil. H_2SO_4, syrupy H_3PO_4, diphenylamine indicator.

Principle:

In acid medium, dichromate oxidises Fe^{2+} to Fe^{3+}, while $Cr_2O_7^{2-}$ is reduced to green Cr^{3+}. Diphenylamine (internal indicator) turns from green to blue-violet at the end point. H_3PO_4 is added to form a colourless complex with Fe^{3+}, which lowers the Fe^{3+}/Fe^{2+} potential and makes the colour change sharp.

$$Cr_2O_7^{2-} + 14\,H^+ + 6\,Fe^{2+} \rightarrow 2\,Cr^{3+} + 6\,Fe^{3+} + 7\,H_2O$$

Procedure:

1. Fill the burette with standard 0.1 N $K_2Cr_2O_7$ solution.
2. Make the given ferrous solution up to the mark in a 250 mL standard flask.
3. Pipette 20 mL of it into a conical flask; add 20 mL of dil. H_2SO_4, 5 mL of syrupy H_3PO_4 and 2–3 drops of diphenylamine indicator.
4. Titrate with $K_2Cr_2O_7$ until the green colour changes to blue-violet (permanent).
5. Repeat to obtain concordant readings.

Observation table:

S.No	Vol. of Fe^{2+} (mL)	Burette reading (mL)	Vol. of $K_2Cr_2O_7$ (mL)
1	20		
2	20		
3	20		

Calculation:

$N_1V_1 = N_2V_2 \rightarrow N_{Fe2+} = (0.1 \times V_{K2Cr2O7}) / 20 =$ _____ N

Amount of Fe^{2+} = Normality x Eq. wt. (55.85) = _____ g/L = _____ g in 250 mL

Result: Amount of ferrous iron in the given solution = _____ g/L.